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REVIEW

Artificial Intelligence in Landscape Ecology: Recent Advances, Perspectives, and Opportunities

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Abstract

Purpose of Review Artificial intelligence (AI) is disrupting science and discovery across disciplines, offering new modes of inquiry that are changing how questions are asked and answered and upsetting established norms. In this paper, we review the state of the art of AI in landscape ecology and offer six areas of opportunity for landscape ecologists to capitalize on AI tools moving forward. These areas include geospatial AI (GeoAI), geometric AI, Explainable AI (xAI), generative AI (GenAI), Natural Language Processing (NLP), and robotics.

Recent Findings Landscape ecology has a long history of using AI, notably machine learning methods for image classification tasks, agent-based modeling, and species distribution modeling but also knowledge representation and automated reasoning for landscape generation and spatial planning. Methods have become more diverse and complex in recent years, with a new generation of AI-based tools rapidly emerging. These new tools have potential to improve how landscape ecologists map, measure, and model landscape patterns and processes as well as improve the explainability of model outputs.

Summary There are many untapped opportunities for landscape ecologists to leverage emerging AI-based tools in research and practice including generating virtual landscapes for simulating processes such as wildfires and leveraging natural language processing to generate new insights from text data. Regardless of the application, researchers using AI tools must also consider the ethical implications of data and algorithmic biases and critically assess how these methods can be used responsibly.

Keywords Interpretable AI · Landscape digital twins · Deep SDMs · Ethics · Data management

Introduction

The use of Artificial Intelligence (AI) has exploded across academia and industry, disrupting how science is approached and the types of questions that can be asked and answered. However, these new modes of inquiry are also upsetting established norms for testing the accuracy and validity of findings and unsettling trust in outcomes through data management issues and a lack of transparency in the underlying systems. As AI-based methods make their way into mainstream science and practice, it is an opportune time to review how these technologies are being used in landscape ecology and discuss areas where the newest generation of tools might be leveraged further to improve predictions and interpretations so that researchers can make informed decisions about integrating these tools in their own analyses.

The term "artificial intelligence" was first coined in 1955 as part of a proposal for a summer workshop held at Dartmouth College, in New Hampshire, USA. The aim of the workshop was two-fold: first to explore the theory that human intelligence could be precisely described by a machine in such a way as to be simulated, and second, to understand how to make machines use language, form abstract concepts, and solve problems that, up until then, had been reserved for humans [1]. From that initial start, the field of AI focused on creating knowledge related to the development of complex cognitive models, the simulation of human information-processing, and overall understanding of human intelligence [2]. However, at the same time, the field was also branching into a parallel, and sometimes colliding, tract focused on evoking intelligent behavior from machines, regardless of whether the computer process

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followed plausible human cognitive mechanisms [2]. This latter track has morphed into the popular form of 'AI' today, where it is accepted that although machines can achieve similar outcomes as humans achieve with intelligence [3], the method by which computers achieve these outcomes is very different than how humans process information [4].

While there is no singular definition of the concept today, AI can broadly be considered the study of agents that receive percepts from the environment and perform actions based on those percepts [5]. Learning can then extend the reach and performance of the agent into unknown environments. Applied across the sciences, the term AI is widely used to refer to the category of technologies that leverage these types of agents for data analytics, predictions and forecasting, object categorization, natural language processing, and other tasks. While machine learning, which refers to how machines learn and improve from experience without being explicitly programmed, has become a centerpiece of many AI-based tasks, it is just one of a suite of AI technologies that are often presented as a nested hierarchy (Fig. 1). Approaches like neural networks and deep learning are rooted in machine learning but use different approaches to simulate decision making, while techniques like natural language processing and automated reasoning are considered in a different sphere.

The field of landscape ecology has been engaging with AI for many decades, mostly through the use of machine learning-based algorithms but also through knowledge representation and automated reasoning. Machine-learning based techniques for image classification (e.g., maximum likelihood classifiers) have been in use since the early 1970s [6], and certain types of agent-based models (ABMs) as well as some species distribution modeling (SDM) frameworks such as Maxent [7], operationalize machine learning to support landscape ecology investigations. However, the newest wave of AI is advancing disciplinary frontiers by enabling the processing and analysis of vast datasets with remarkable speed and precision. As a quantitative and computational science, landscape ecology has an opportunity to leverage these new tools to uncover new knowledge through the predictive understanding of complex patterns that would otherwise remain hidden. However, it is important that these technologies are applied responsibly and ethically so that

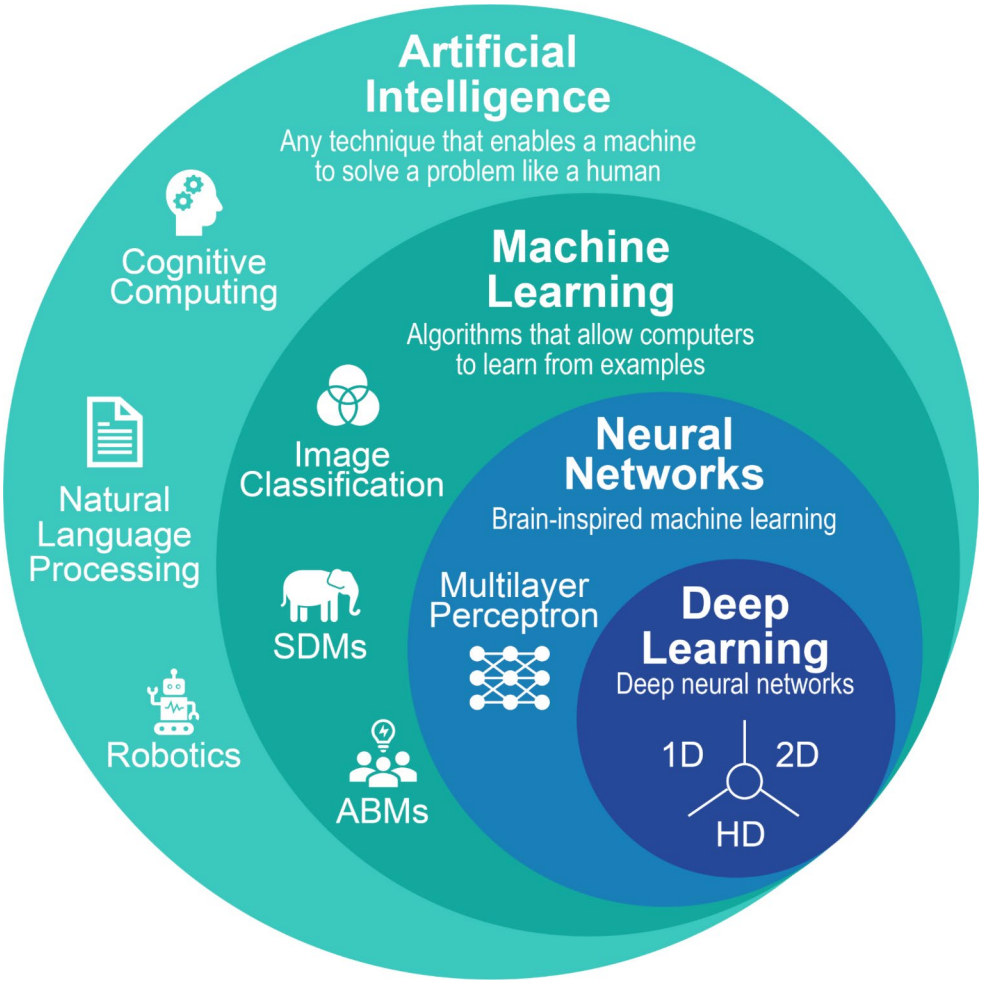

Fig. 1 The nested hierarchy of artificial intelligence (AI), machine learning (ML), neural networks, and deep learning (DL), although boundaries are sometimes fuzzy in terms of which techniques fit into which layers. The examples provided are techniques frequently used in landscape ecology, which include certain types of image classification methods, species distribution models (SDMs), and agent-based models (ABMs). Deep learning can further be partitioned into 1-dimensional (1D), 2-dimensional (2D), or higher dimensional (HD) tasks

they do not limit creativity, create more uncertainty, or perpetuate inequalities.

We review the ways in which AI has contributed to landscape ecology advances in the past and how it can be further leveraged to uncover new insights into the patterns and processes of landscapes. Many uses of AI in landscape ecology have been through machine learning and the sub domain of deep learning. However, advances have also been taking place in areas such as knowledge representation, heuristics, and automated reasoning. We first review past and recent advances in landscape ecology that have used AI-based techniques, including certain agent-based models (ABMs), species distribution models (SDMs), computer vision and pattern recognition, landscape generation and spatial prioritizations for planning. We then shift our discussion to look ahead and offer perspectives on six areas where the latest wave of AI advances can contribute to advancing landscape science and discovery moving forward. These areas include explainable AI (xAI), generative AI (GenAI), natural language processing (NLP), geospatial AI (GeoAI), geometric AI, and robotics. We conclude by discussing some of the ethical responsibilities that researchers must consider when using AI tools for research.

Past and Present Uses of AI in Landscape Ecology

The field of landscape ecology has a long history of incorporating AI-based methods that stems from its computational roots [8] and tight coupling with GIS, geocomputation, and remote sensing [9]. Early on, machine learning algorithms were used both implicitly to classify remote sensing images into land cover maps (e.g., through maximum likelihood, support vector machine, or random forest techniques), and more explicitly through modeling and prediction tasks, such as for agent-based modeling and species distribution modeling (e.g., MaxEnt), among others. While these methods have been widely used in the field for decades, there has been an uptick in recent years, with methods getting more diverse and approaches more complex over time [10]*.

Perhaps some of the earliest explicit applications of AI-based methods in landscape ecology were through agent-based models (ABMs), which are simulations for studying the interactions between people and the environment. In ABMs, autonomous agents are programmed through rules to behave and interact, and the agents can change their actions based on feedback from prior decisions [11, 12], producing emergent properties that may differ from individual actions alone. ABMs were employed in the early 1990s to simulate land use decisions [13], and their use has been reviewed extensively in landscape ecology, coupled human-natural systems, and ecosystem management [14–17]. While not all ABMs are based on AI or machine learning techniques, recent AI-based advances may aid with improving the computational architectures to represent behaviors and learning processes to create more 'intelligent' agents and improve outputs [17]*. These advances are helping to overcome some of the noted limitations of ABMs including their lack of predictive power and transparency as well as validation challenges.

AI-based techniques, notably those in the machine learning realm, have also advanced landscape ecology through species distribution models (SDMs; also called range models or ecological niche models). SDMs predict how species distributions will vary in space and time based on modeled relationships between occurrences and environmental variables. While correlative SDMs have been in use since the early 1980s, within the last two decades the field has moved to more complex statistical models that rely on machine learning, such as the maximum entropy (MaxEnt) framework [18, 19]. These advances have improved accuracy, but machine learning-based models are prone to overfitting, which affects interpretability [20]. 'Deep SDMs' based on neural networks are emerging to better address the complexity of species ranges/niches [21] as they can integrate landscape patterns and heterogeneity at different scales into predictions [22], but interpretability of the outputs still remains an issue. Explainable AI (xAI, discussed below) has the potential to contribute here to decipher the relative importance of predictor variables or their causal effects on focal species [23]**.

The use of deep learning has also gained traction in landscape ecology for ecological monitoring and data mining tasks. For example, a limitation for species distribution modeling is the availability of observations. Camera traps have automated some data collection tasks, but the vast numbers of images captured creates new bottlenecks for users to extract useful information from hundreds or thousands of images [24]. Deep learning-based computer vision techniques are being employed to quickly identify and classify species from images [25] and even detect morphological traits or animal behaviors [26]. These tasks typically use convolutional neural networks (CNNs), which leverage spatial context to identify objects in high-resolution remotely sensed imagery [27] and differ from simpler artificial neural networks (ANNs) in their larger number of hidden layers [28]* and more complicated architecture [29]**. CNNs are being applied to high resolution satellite or drone images to detect large marine animals [30, 31] and locate and track migrating herds [32] and can be used with community science apps like iNaturalist or eBird to help automatically identify species from photographs. Computer vision techniques have also been applied to landscape photographs to

glean insights on human-nature interactions [33] and have the potential to be used to assess attributes such as cultural ecosystem services, landscape aesthetic value, preferences, among other characteristics. The use of online, AI-based search engines such as Google Image Search is also gaining traction for mining landscape photos from massive collections according to search terms [34].

A sophisticated array of deep learning methods are also being used in landscape ecology to analyze acoustic data and audio recordings for detecting species and monitoring their welfare and habitat health [35, 36]. These models have demonstrated enhanced ability to parse high volume acoustic data even through noisy signals [37, 38], which is facilitating pioneering work in decoding animal communication systems and contributing to a deeper understanding of animal behavior at the landscape level [39]. Continued progress in this field will depend on the availability of datasets and ongoing development of soundscape theories in landscape ecology [39, 40].

Beyond machine learning, the field of landscape ecology has been advancing through knowledge representation and automated reasoning for virtual landscape generation and spatial planning. Generating neutral landscapes to simulate processes and test hypotheses has been a cornerstone of landscape ecology with many stand-alone programs such as Rule [41], QRule [42], and SIMMAP [43], and software packages [44–46] being developed. However, the early algorithms for these models had limited flexibility, partly because certain parameters did not have direct landscape metric analogs [47] or could not be reverse engineered from a real-world map [48]. Using AI, researchers have started to overcome these limitations by structuring information in a way that makes it easier for computers to solve the problem. For instance, Slager [49] developed an AI search method based on heuristic optimization and a multi-objective evolutionary algorithm that can generate landscape configurations in a plausible, generic manner. The method has been adopted for creating spatial plans [50] and used to develop software for landscape generation based on user-defined composition and configuration values corresponding to common landscape metrics [51]. The models have been improved for scalability [52] using constraint programming, which is an AI technique based on automated reasoning, via the Choco-solver [53].

Spatial planning is another area where there has been solid adoption of AI-based methods to improve prioritization tasks for many years. The popular Marxan software utilized metaheuristics through simulated annealing for spatial prioritization [54]. Later, integer linear programming was demonstrated to improve the quality of solutions and processing time [55], and forms the basis for the widely-used prioritizr R software package [56]. Constraint programming has been shown to provide both constraint satisfaction and optimality guarantees for spatial conservation planning optimizations [57]. More recently, the conservation area prioritization through artificial intelligence (CAPTAIN) software uses reinforcement learning, a form of machine learning used to optimize spatial planning results [58].

Opportunities for Landscape Ecology to Leverage Emerging AI Tools

While machine learning, knowledge representation, automated reasoning, and other AI-based methods have been in use in landscape ecology for many decades, a new generation of powerful AI tools is emerging that have the potential to be leveraged effectively to advance how landscape ecologists map, measure, and model landscape patterns and processes as well as improve the interpretability of model outputs. In this section, we offer perspectives on six emerging areas of AI tool development and discussions on how landscape ecologists can both use and contribute to these advancing areas (Fig. 2). These areas include: (1) Explainable AI (xAI); (2) Generative AI (GenAI); (3) Natural Language Processing; (4) Geospatial AI (GeoAI); (5) Geometric AI; and (6) Robotics.

Explainable AI (xAI)

A common complaint of many machine learning-based models, such as the SDMs used in landscape ecology, is that they are 'black box', meaning it is difficult to interpret how the model determined its output. Explainable AI (xAI), also sometimes called Interpretable AI, aims to decipher the behavior of complex models to uncover the reasons why or how an AI-based model determined its response. In other words, xAI aims to generate more transparent and understandable results. xAI can be applied to many types of AI techniques. In landscape ecology, xAI techniques are promising for generating more predictable species distribution models [23]**. As the algorithms driving SDMs have become increasingly complex, both mathematically and computationally, predictive accuracy has also increased, but these gains have come at the cost of interpretability [59], making SDMs a promising area for xAI in landscape ecology.

xAI methods are categorized based on whether they target a direct understanding of the model architecture (i.e., model-based interpretability) or whether they analyze post-hoc model behavior (i.e., after model fitting) [60]. Model-based xAI methods integrate explainability directly within the architecture [61] by considering intermediate semantic concepts that are relevant to the final prediction. They thus

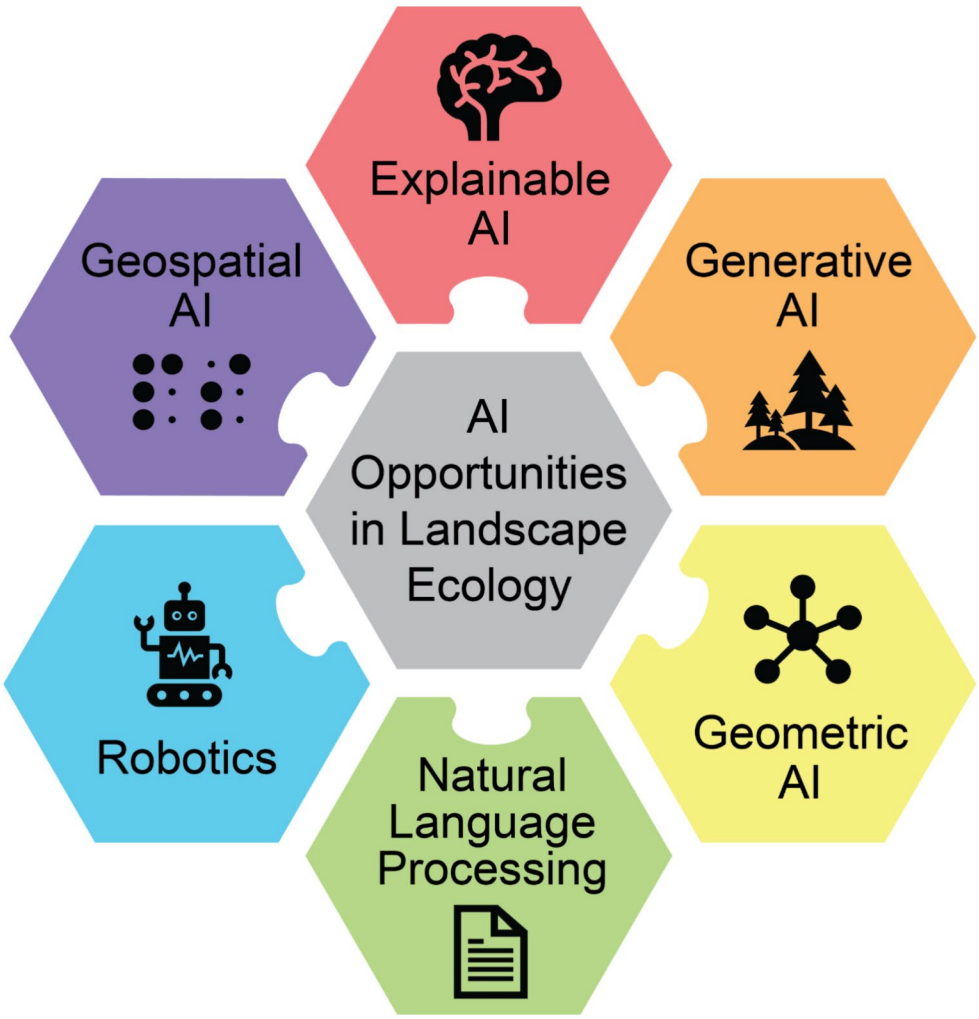

Fig. 2 Six opportunity areas for landscape ecology to leverage the newest generation of advances in AI to uncover new knowledge through predictive understanding of complex patterns that would otherwise remain hidden

have the potential to make deep learning models in landscape ecology more interpretable by leveraging a priori knowledge to incorporate ecological processes explicitly. For example, to predict forest cover from remote sensing images using a CNN model, tree height and canopy density can be incorporated as intermediate variables into the model structure to better explain forest types [62]. However, model-based interpretability methods demand significant prior knowledge, which can limit their generalizability.

Post-hoc xAI methods are more versatile, allowing comparisons and integrations across models with varying architectures. Two popular methods for post-hoc interpretability are Shapley values and local interpretable model-agnostic explanation (LIME). Shapley values [63]) were developed in cooperative game theory to understand how to fairly distribute the 'payout' of a game amongst the different players. Translated into landscape ecology, Shapley values allow a prediction (e.g., from an SDM) to be treated as the 'payout' with the goal to understand how much each player (i.e., each environmental predictor in the model) contributed to the payout (i.e., the prediction). Using a form of replacement bootstrapping, the marginal contribution of each variable can be determined. An extension to Shapley has been developed to explain individual predictions called SHapley Additive exPlanations, or SHAP [64]. Both Shapley values and SHAP have recently begun to be used in ecology and landscape ecology for explaining the contributions of environmental drivers of species richness, habitat conditions, and ecosystem state transitions, among others [65–67] and are a promising area for future research.

The second method, LIME [68], approaches post-hoc model explanation by training an interpretable model locally around a prediction based on perturbations of the original data in order to test how the predictions change when a model is fed variations of the data. The interpretable model is a close approximation of the machine learning model, so predictions made locally can be compared across models to explain the prediction. Both LIME and Shapley values/SHAP offer a way to generate more transparent and understandable predictions in landscape ecology. However, neither can directly answer questions about causality because modeled associations are not equivalent to real causal

associations [23, 69]. There is still much work to be done linking these xAI tools with ecological questions. In particular, the use of xAI to understand the role of climatic drivers on current and future species distributions is needed to plan for impending biodiversity shifts with climate change. Since both Shapley/SHAP and LIME grow more computationally intensive with increasing input variables, they may not be suitable for models with many inputs.

Generative AI (GenAI)

Generating neutral landscapes for hypothesis testing has long been a focus of landscape ecology, and as reviewed above, researchers have been drawing on AI-based techniques including heuristics and automated reasoning to improve these models for several years. Generative AI (GenAI) is another way in which landscape generation can be approached. GenAI models learn patterns and structure from training data to create new content, such as text, images, simulations, or videos. Leveraging GenAI along with AI-based solvers, researchers can create simulated, three-dimensional (3D) landscapes that have been trained on real-world data. Generated 3D landscapes could be used to simulate disturbances such as wildfires or pest outbreaks, test interventions, and run counterfactuals. Early attempts suggest GenAI can be used for global change studies such as forecasting the effects of climate change on the spatial features of landscapes [70]. There are valid concerns over how GenAI technology is being used to create 'deepfakes', including falsified satellite imagery [71], however there is also vast potential in using GenAI to create realistic, 3D landscapes that can be used to test hypotheses and develop new measures of landscape pattern and process.

GenAI may also be useful for structuring inputs and synthesizing outputs of landscape digital twins. A digital twin is a set of virtual information constructs that mimics the structure, context, and behavior of a natural, engineered, or social system (or system-of-systems). Ideally, a digital twin is dynamically updated with data from its physical system twin, has a predictive capability, and can inform real-world decisions. Digital twins have most often been developed and deployed in closed systems, like manufacturing, but are becoming more established in urban systems, where questions of spatial relationships and accuracy are starting to be debated [72]. Their use in other landscapes remains aspirational, primarily because environmental systems present challenges such as non-linear dynamics. GenAI has the potential to streamline digital twin development and deployment in these environments where the landscape system provides training and validation data for GenAI in a virtual 'test and learn' environment, where verification, validation, and uncertainty are also measured (Fig. 3). In landscape ecology, coupling GenAI and digital twins could help build scenarios of real-world situations and potential outcomes to offer early warnings of impending disturbances, and provide a testbed on which to test interventions (e.g., forest restoration or prescribed burning) and learn the outcomes before actually changing the landscape. Since some landscape interventions present high risk, there is the potential for large payoffs in developing these types of GenAI-aided digital twins with human-in-the-loop decision making.

Natural Language Processing

Landscape ecology, while firmly rooted in quantitative traditions, also involves qualitative analyses to understand people's motivations, attitudes, and beliefs about land and seascapes, and studies often blend quantitative and qualitative methods to analyze social-environmental systems [73]. Natural language processing (NLP) is a form of AI

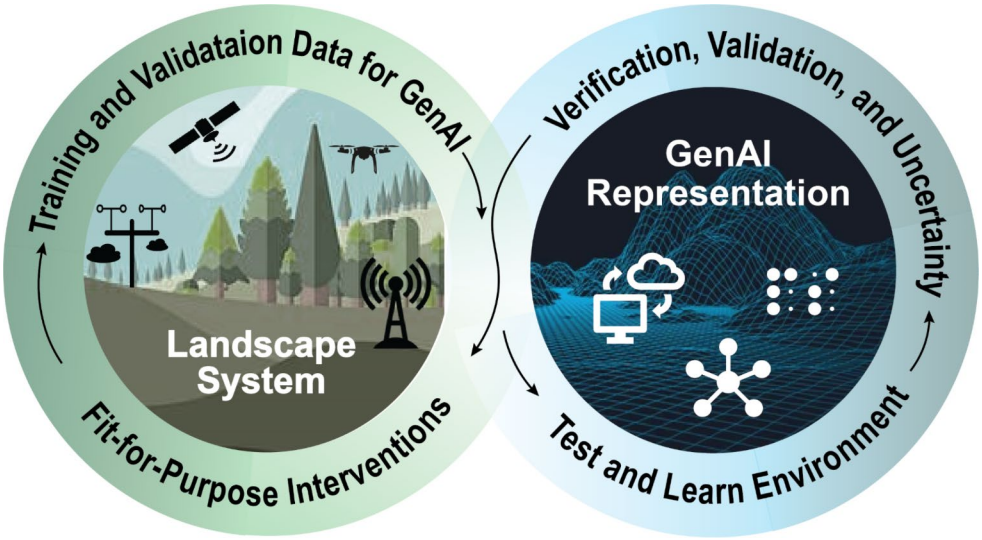

Fig. 3 A Digital Twin that couples training and validation data from the physical landscape system with Generative AI (GenAI) to drive the virtual representation in a test and learn environment that also contributes verification, validation, and uncertainty before fit-for-purpose interventions are performed in the physical system to minimize costs and risk

that gives computers the ability to interpret, manipulate, and comprehend human language from sources such as text, audio, social media posts, or news feeds [74]. The most well-known example of natural language processing is ChatGPT. While ChatGPT was designed to respond to user inquiries, other forms of natural language processing can help summarize or synthesize large amounts of text. In landscape ecology, studies often rely on social media posts or other volunteered content to capture aspects of the landscape that cannot be gleaned from quantitative data, such as remote sensing. Natural language processors can be leveraged to distill those forms of big data into interpretable summaries. Studies have started using NLP to assess social media posts related to urban park experiences [75] and have coupled this technology with landscape metrics to better understand how people experience landscapes [76].

Geospatial Artificial Intelligence (GeoAI)

GeoAI integrates geospatial studies and AI for studying Earth-related problems. GeoAI includes both the application of AI methods within geography and the geosciences and the incorporation of spatial, temporal, and place-based dimensions into AI methodologies [77]. This integration is particularly beneficial for landscape ecology, which is an inherently spatial science that is also focused on place-based analyses. CNNs (introduced above) have been widely adopted in landscape ecology due to their ability to learn hierarchical features directly from images, which is crucial for accurately analyzing spatial patterns in landscape ecology. As GeoAI continues to evolve, including through innovations in pattern recognition and transformer models, there is the opportunity for applications to extend into more complex, high-dimensional analyses, such as time series (e.g., long-term climate impact studies) and 3D landscape structure modeling [78]. In particular, 3D predictive models have the potential to offer a nuanced understanding of the vertical spatial structure of habitats.

Several emerging innovations in GeoAI have the potential to change how landscape data is modeled and interpreted. Neuro-Symbolic AI, which combines neural networks with more traditional logic and predefined rules for knowledge representation, reasoning, and decision-making, can be used in GeoAI to interpret complex satellite or drone images and then answer questions about the image's content or to infer relationships between objects. Similarly knowledge graphs, which organize data and forge connections across multi-scale or multi-source data can be helpful for analyzing complex relationships and interactions between data points, or locations, which are common in human-natural systems. While still developing, these methods can provide scalable and interpretable AI models that enhance decision-making processes in landscape ecology [79]. However, building and fine-tuning these models is both time-consuming and energy-intensive. As spatial data become more abundant, a future focus should be creating a pool of foundation models that are transparent, explainable, and incorporate physical mechanisms to enhance their interpretability and applicability [39, 40]. These models could streamline the process of adapting AI to specific landscape ecology contexts, reducing the need for extensive retraining while ensuring that predictions remain grounded in ecological principles.

Geometric AI

A limitation of GeoAI in landscape ecology is that most approaches rely on regular grid structures for landscape analysis, despite the fact that many relationships in landscape ecology can be modeled as non-Euclidean graphs where proximity is represented by genetic, organizational, institutional, social, or other conceptualizations of proximity [80]. Researchers are seeking to overcome this limitation by including geometry and spatial autocorrelation explicitly in AI-based models [81, 82]. Graph-based deep learning (GDL), particularly through graph neural networks, extends traditional deep learning techniques to non-Euclidean domains by leveraging graph structures and incorporating more complex geometric structures, such as manifolds and 3D meshes [29]. These additions make Geometric AI a powerful tool for analyzing the irregular data structures and complex spatial relationships that are often found in landscape ecology. For instance, GDL introduces innovative approaches for modeling and analyzing connectivity between landscape elements, such as habitat patches, by incorporating high-dimensional and non-spatial data. GDL extends traditional graph-based methods by enabling landscape structures to be projected over non-Euclidean spaces (e.g., topographical surfaces) or modeled directly in 3D or higher dimensions. Unlike static resistance-based methods, such as circuit theory, GDL can dynamically integrate both ecological and external factors (e.g., political boundaries, land use policies) into a unified framework. Additionally, GDL is scalable to large volumes of data, making it more adaptable for modern ecological datasets, which often involve complex interactions across various dimensions and scales.

The emerging field of Geometric AI holds significant potential for advancing landscape ecology research by effectively capturing spatial dependencies, modeling complex landscape networks, and predicting ecological patterns in ecologically meaningful ways. In turn, the expertise of landscape ecologists, particularly in understanding ecological theories and spatial structures, can provide valuable insights that contribute to the development and refinement

of graph-based deep learning methodologies. However, these models are often structurally complex, which can make them computationally intensive and challenging to interpret and generalize. Additionally, these models require high-quality datasets that accurately capture the complexity of geometric structures [29, 83]. In landscape ecology, obtaining such datasets is particularly challenging due to the variability and dynamic nature of ecosystems, as well as the difficulty in obtaining detailed 3D data over large spatial extents. Addressing these challenges and adapting graph-based deep learning models to the specific needs of landscape ecology research will require close interdisciplinary collaboration between computer scientists and ecologists.

Robotics

Robotic systems, especially drones and other types of autonomous vehicles, are poised to revolutionize the way in which species and landscape data are collected. Since autonomous vehicles can access remote or other areas that are challenging to reach physically, they have the potential to fill in missing data gaps while also protecting sensitive habitats from disturbance. To this point, autonomous vehicles have been used primarily for capturing high resolution imagery [84, 85], from which land cover and other landscape variables can be extracted. However, image collection is just one of the many functions of drones, and other capabilities are currently underutilized in landscape ecology. For instance, drones can be outfitted with customized sampling equipment, and researchers are experimenting with robotic systems that can cut and sample leaves and other plant material [86, 87] or capture specimens with sticky paper and other devices affixed to the platform [88]. Fully autonomous systems can learn from the data they are collecting in real time, which can lead to real-time decisions or changes in sampling strategy. Amphibious or underwater drones, like the 'SnotBot', which has been engineered to collect whale and dolphin blow on petri dishes analyses of marine health (Keller and Willke 2019), can aid in advancing a more holistic and coupled land and seascape ecology [89].

Concluding Remarks

AI has the potential to revolutionize how research is conducted in landscape ecology, however, there are issues that complicate the use of AI including data representativeness, reproducibility, and ethics. First, we know there is bias in published applied ecology and conservation studies toward the Global North and wealthy countries [90–92], which means that any use of AI trained on these studies, or the data they generated, may also inherent similar biases. With biased training data, AI-generated landscapes might not be reflective of certain ecosystems, cultures, or people, and natural language processors may miss key perspectives and viewpoints. While it is ultimately the responsibility of editors and publishers to determine the content published in peer-reviewed journals, the onus shifts to researchers to understand the biases inherent in the data they are using with AI and ensuring the biases that already exist in society are not replicated in a manner that lends false scientific credibility.

Second, ensuring transparency and reproducibility is crucial for AI-based advances to be both impactful and ecologically robust [10, 29, 93], particularly since AI-based models can be sensitive to variability in data, hyperparameters, or other settings. Minimally, every stage of the research process should be documented including explicit articulation of hypotheses and detailed descriptions of the methods. It is also essential to document the computational environment in which the code was run, record the final configurations of the workflow, and provide information for future use, such as training duration, fine-tuning requirements, sensitivity to hyperparameters, and hardware specifications [94]. Lastly, permanent access to code and data should be maintained (e.g., through a DOI) so that anyone can reproduce the findings.

Ethically responsible use of AI in landscape ecology extends beyond issues of data bias and reproducibility to understanding how these tools may upset privacy norms, how future decisions made with AI might impact the land and humans, and how accountability should be shared amongst the human actors. The topic of landscape digital twins provides a launch point for this discussion. Development of a landscape digital twin necessarily involves the placement of sensors, cameras, and recording devices, which may violate privacy of those moving through the space. Since certain landscapes can be considered a 'commons' resource [95], decisions about their management should benefit society and minimize harm. If the digital twin is being used to make intervention decisions or to allocate or extract resources, then it is critical that any cost-benefit tradeoffs are considered during the process. Lastly, if there are unjust, unfair, or adverse consequences of an intervention made based on AI, it is important to consider who should reasonably share responsibility for that decision. Is it the person who designed the sensor network in a manner that captured geographically biased data? Is it the person who programmed the algorithms to learn on biased data? Is it the person who initiated the intervention based on flawed findings? These are just some of the ethical questions that landscape ecologists and others will need to grapple with as AI permeates scientific processes.

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  This article calls for urgency in converging the research paradigm between ecology and AI and the need for more purposeful synergy across both fields to accelerate advances.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing Interests Amy Frazier and Lei Song declare they have no conflict of interest to disclose.

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